

# vennu

every venue · every menu

## Product Development Roadmap

Version 4 · Full Platform Specification

16

Phases

6

Display

Layouts

3

POS

Systems

5

TV

Platforms

30+

Feature

Flags

React 18 + TypeScript · .NET 8 Web API · ASP.NET SignalR · EF Core 8 · Azure SQL + Blob

Square · Toast · Clover · Claude AI · Android TV · Samsung Tizen · LG webOS · React Native

# Platform Overview

Vennu is a cloud-connected digital signage platform built specifically for restaurants, bars, breweries, cafes, and food halls. Unlike generic signage tools it is designed from the ground up for hospitality — with real-time POS integration, restaurant-native display layouts, multilingual support, and a staff mobile app that lets a solo operator update their menu in 30 seconds from the kitchen. The tier system and non-invasive upgrade prompts turn the platform into a self-serve revenue engine that grows with every venue.

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| Backend     | .NET 8 Web API · ASP.NET SignalR · EF Core 8 · Azure App Service         |
| Database    | Azure SQL · Azure Blob Storage · Azure SignalR Service                   |
| Frontend    | React 18 + TypeScript · Vite · react-i18next · @microsoft/signalr        |
| Real-time   | SignalR WebSocket hub · ~200ms push · Service Worker offline cache       |
| POS         | Square · Toast · Clover · IPosProvider abstraction                       |
| TV / Mobile | Browser · Android TV · Fire TV · Samsung Tizen · LG webOS · React Native |
| AI          | Claude API (Anthropic) · descriptions · translation · custom displays    |
| Tiers       | \$39 Starter · \$49 Restaurant Starter · \$89 Pro · \$179 Business       |
| HaaS        | \$89–249/mo bundles · hardware ownership transfers at contract end       |

## All 16 Phases at a Glance

|                                      |                                     |                                          |                                       |                                      |                                       |                                        |                                      |
|--------------------------------------|-------------------------------------|------------------------------------------|---------------------------------------|--------------------------------------|---------------------------------------|----------------------------------------|--------------------------------------|
| 01<br>Foundation<br>Weeks 1–3        | 02<br>Core Backend<br>Weeks 3–7     | 03<br>Tier System<br>Weeks 6–9           | 04<br>Admin CMS<br>Weeks 8–12         | 05<br>Display Layouts<br>Weeks 10–14 | 06<br>Display Layouts<br>Weeks 13–17  | 07<br>Scheduling Engine<br>Weeks 15–18 | 08<br>Tap List Boards<br>Weeks 17–21 |
| 09<br>Upgrade Prompts<br>Weeks 19–23 | 10<br>POS Integration<br>Months 5–8 | 11<br>Multilingual Support<br>Months 7–9 | 12<br>Staff Mobile App<br>Months 9–11 | 13<br>TV Apps<br>Months 11–15        | 14<br>Super Admin CRM<br>Months 13–16 | 15<br>AI Features<br>Months 14–18      | 16<br>Analytics<br>Months 17–22      |

0  
1

# Foundation & Launch Setup

Minimal viable setup to start building immediately. Azure infrastructure at dev scale costs under \$20/mo. Business accounts open. Tooling configured. First lines of code can be written by the end of week one.

[ SETUP ]

Phase 01

Duration: Weeks 1–3

Status: SETUP

Milestone: Ready to build

## Azure Infrastructure — Dev Tier

- |                                            |                                                                       |
|--------------------------------------------|-----------------------------------------------------------------------|
| ■ <b>App Service B1 · \$13/mo</b>          | Dev/test tier · scales to B2 when first customer signs up             |
| ■ <b>Azure SQL Basic · \$5/mo</b>          | 2GB · sufficient for hundreds of venues at launch                     |
| ■ <b>Azure Static Web Apps · Free</b>      | CDN-backed hosting for all React SPAs globally                        |
| ■ <b>Azure Blob Storage · ~\$1/mo</b>      | Menu photos, tap photos, custom display assets, HTML templates        |
| ■ <b>Azure SignalR Service · Free tier</b> | Up to 20 concurrent screen connections · upgrade to Standard at scale |
| ■ <b>GitHub repo + Actions CI/CD</b>       | Auto-deploy to Azure on every push to main · zero manual deployment   |

## Business Setup

- |                                              |                                                                        |
|----------------------------------------------|------------------------------------------------------------------------|
| ■ <b>Stripe account</b>                      | 2.9% + 30c/transaction · subscriptions, trials, invoicing all built in |
| ■ <b>Business bank account</b>               | Mercury or Relay · free · keeps business finances cleanly separated    |
| ■ <b>Domain — venuu.app</b>                  | Register immediately · \$12/yr · also check getvennu.com as backup     |
| ■ <b>Trademark 'Venuu' — USPTO Class 042</b> | File this week · \$250–350 + attorney · foundational brand protection  |

| Dev Tooling                    |                                                                                |
|--------------------------------|--------------------------------------------------------------------------------|
| ■ Visual Studio 2022 Community | Free · full .NET 8 IDE with EF Core scaffolding and SignalR templates          |
| ■ Vite + React 18 + TypeScript | Lightning-fast build tool · HMR in milliseconds · same setup across all SPAs   |
| ■ Postman                      | Free · API testing and webhook simulation · import collections per integration |
| ■ Figma · \$15/mo              | UI design · prototypes already exist from planning phase · now refine them     |

02

Core Backend & Real-Time Engine

Build the .NET 8 API and SignalR hub that every feature in every later phase depends on. Get screen registration, the heartbeat, and real-time push working end-to-end before building any UI. A change in the admin should appear on a TV within 200ms.

[ BUILD ]

Phase 02

Duration: Weeks 3–7

Status: BUILD

Milestone: Screens  
update in real time

Core Data Models — EF Core 8

|                                 |                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------|
| ■ Venue                         | Top-level tenant · name, timezone, type, branding, subscription                 |
| ■ Screen                        | Physical display · screenKey, wallGroup, position, lastSeen, status             |
| ■ Menu / MenuSection / MenuItem | Full hierarchy · price, hhPrice, availability, qty, tags, imageUrl              |
| ■ MenuItemTranslation           | (ItemId, LanguageCode, Name, Desc) · multilingual baked in from day one         |
| ■ TapItem                       | Beer-specific model · name, style, ABV, IBU, description, glassColor, nameColor |
| ■ ScreenPairingCode             | 6-digit code · venueId, screenId, expiresAt, isClaimed · used for TV setup      |

REST API — Core Endpoints

|                                          |                                                                 |
|------------------------------------------|-----------------------------------------------------------------|
| ■ POST /api/venues                       | Create venue · assigns default tier                             |
| ■ GET /api/venues/{id}/content           | Full board payload for a venue · used by admin preview          |
| ■ GET /api/display/{screenId}/content    | Full board payload for a screen · called by display app on load |
| ■ POST /api/display/{screenId}/heartbeat | Screen health ping every 30s · updates lastSeen timestamp       |

- **POST /api/screens** Register new screen · returns screenKey (sc-a3f9bc)
- **POST /api/screens/pairing-code** TV calls this · generates 6-digit code with 10-min expiry
- **GET /api/screens/pairing/{code}/status** TV polls this every 3s until claimed
- **POST /api/screens/pairing/{code}/claim** Admin claims code · links screen to venue · TV redirects
- **POST /api/media/upload** Upload image → Azure Blob → returns CDN URL

### ASP.NET SignalR Hub

- **JoinScreen(screenId)** TV connects and joins its named group on startup
- **JoinVideoWall(wallId, position)** Wall screens join shared group · position used for CSS offset
- **ContentUpdated event** Pushed to screen group within ~200ms of any admin save
- **ThemeUpdated event** Pushed when theme builder settings change
- **ItemAvailabilityChanged event** Pushed when POS webhook fires · board patches single item in <500ms
- **SyncTick(serverTimeMs)** 16ms broadcast for video wall frame sync · ~60fps
- **withAutomaticReconnect()** Client reconnects silently after network drops · no manual refresh

### Display Player — Boot Sequence

- **Step 1 — Fetch content** GET /api/display/{screenId}/content · sets initial board state
- **Step 2 — Connect SignalR** Join screen group · listen for ContentUpdated, ThemeUpdated, ItemAvailabilityChanged
- **Step 3 — Start heartbeat** POST /heartbeat every 30s · marks screen Online in admin dashboard

#### ■ Step 4 — Register Service Worker

Caches last payload · board keeps showing if internet drops during service

#### ■ Layout switch

board.layout field determines which React component renders · new layouts = new components only

#### ■ Video wall mode

If wallPosition !== null · full canvas rendered with `translateX(-pos × 1920px)`

### Background Services

#### ■ ScheduleEvaluator

IHostedService · evaluates schedules every 60s · pushes ContentUpdated via SignalR

#### ■ HeartbeatMonitor

Marks screens Offline if no ping in 90s · triggers push alert to admin

#### ■ StripeWebhookReceiver

Handles subscription.updated, invoice.paid, customer.subscription.deleted

## 03

# Tier System & Feature Flags

Build the monetization engine before any customer-facing product features. Every panel, nav item, and button in the Admin CMS (Phase 04) is tier-aware from the first line of code — not retrofitted later. Tiers live in the database, not in code, so new tiers and feature changes take seconds, not deployments.

[ BUILD ]

Phase 03

Duration: Weeks 6–9

Status: BUILD

Milestone: Monetization infrastructure live

## Data Models

|                        |                                                                                   |
|------------------------|-----------------------------------------------------------------------------------|
| ■ Feature              | One row per feature flag · key, label, category, isActive master switch           |
| ■ SubscriptionTier     | Fully dynamic · name, slug, price, maxScreens, isPublic, Stripe product ID        |
| ■ TierFeature          | Join table · which features each tier includes · limitValue for metered features  |
| ■ VenueSubscription    | Links venue to tier · Stripe subscription ID, status, currentPeriodEnd            |
| ■ VenueFeatureOverride | Per-venue unlock or block · reason text · optional expiry date · createdByAdminId |

## Feature Resolution Service

|                                 |                                                                             |
|---------------------------------|-----------------------------------------------------------------------------|
| ■ HasFeatureAsync(venueId, key) | Check override first · fall back to tier · called from every API controller |
| ■ GetFeatureSetAsync(venueId)   | Returns full computed feature set for a venue · used by admin frontend      |
| ■ Override beats tier always    | Most specific wins · enables beta testing, custom deals, grandfathering     |
| ■ Metered features              | limitValue field stores limits like '20' for AI descriptions per month      |



## Initial Tier Definitions

|                                           |                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------|
| ■ Starter · \$39/mo · 2 screens           | Photo Grid, Classic Diner, basic scheduling, allergen badges, basic analytics         |
| ■ Restaurant Starter · \$49/mo · 1 screen | Adds: meal periods, bilingual, AI translation (1 lang), Quick Update, combo builder   |
| ■ Pro · \$89/mo · 6 screens               | Adds: all layouts, happy hour, POS integration, staff app, video wall, full analytics |
| ■ Business · \$179/mo · unlimited         | Adds: AI custom builder, multi-location, white-label, HTML editor, unlimited AI       |
| ■ 14-day free trial · no credit card      | Low friction entry · customer tests on their actual TV before committing              |
| ■ Annual billing · 2 months free          | ≈17% discount · dramatically improves cash flow · target 40–60% of customers          |

## Stripe Integration

|                                   |                                                                           |
|-----------------------------------|---------------------------------------------------------------------------|
| ■ Products + Prices per tier      | One Stripe product per tier · monthly and annual price objects            |
| ■ Subscription lifecycle webhooks | subscription.created → activate · past_due → warn · canceled → restrict   |
| ■ Trial handling                  | trialing status → full Pro features · converts automatically at day 14    |
| ■ Tier change via admin           | Upgrade or downgrade triggers Stripe proration automatically              |
| ■ HaaS contract billing           | Separate Stripe subscription for hardware · buyout charge on early cancel |

04

# Admin CMS — Core Editing

The interface restaurant and bar owners use every day. Built with tier-awareness from the start — every locked feature shows a gentle hint rather than an error. Optimised for the specific workflows of running a venue, not a generic CMS.

[ BUILD ]

Phase 04

Duration: Weeks 8–12

Status: BUILD

Milestone: First venue can manage their board

## Menu Editor

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| ■ Section expand / collapse      | Manage large menus without scrolling · focus one section at a time         |
| ■ Inline item editing            | Edit name, description, price, HH price without leaving the menu           |
| ■ Availability toggle — 86 items | One click marks item unavailable · Live / Off pill · SignalR push in 200ms |
| ■ Sold-out / limited qty badges  | 'Only 3 left!' badge on display · auto-removes when quantity hits zero     |
| ■ Allergen & dietary badges      | GF · V · Vegan ■ · Halal ■ · Kosher ☆ · Nuts ■ · Spice level ■             |
| ■ Bestseller tag                 | ★ Popular badge shown on display board · driven by admin toggle            |

## Quick Update Mode

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| ■ Purpose            | Built for the solo restaurant owner running lunch service alone     |
| ■ Daily special push | Type it, hit push · live on all screens in under 10 seconds         |
| ■ 86 toggle list     | Every menu item visible with one-tap on/off · no section navigation |

**■ Auto-midnight reset**

All 86'd items restore automatically · no morning cleanup required

**■ Mobile-first layout**

Optimised for a phone screen · large touch targets · no desktop needed

**Screen Management — Phase 04 Registration****■ Generate registration URL**

Admin creates screen record · gets  
vennu.app/display/sc-a3f9bc

**■ Screen health dashboard**

Online / Offline status per screen · last-seen timestamp · screen count

**■ Screen naming**

'Main Bar TV', 'Patio Board', 'Entrance Display' · location label

**■ Push to specific screen**

Target one screen or all screens simultaneously

**■ Multi-screen overflow visualiser**

Mini screen strip showing item distribution across screens

**Tier-Aware UI Patterns****■ HasFeatureAsync check on every panel**

Every tab and section checks feature flag before rendering

**■ Locked nav items at 50% opacity**

Features visible but dimmed · tier badge (e.g. 'PRO') shown beside them

**■ Locked section blurred preview**

Glimpse of what the feature looks like · one benefit sentence · soft CTA

**■ HH price field disabled below Pro**

Field visible but greyed with tier badge · no confusion about why it exists

**■ Upgrade modal on locked click**

Bottom sheet · shows feature benefit + all tier features + single CTA

05

# Display Layouts — Restaurants & Cafes

The first two display layouts cover the broadest market: quick-service restaurants, ethnic restaurants, cafes, and diners. Photo Grid is the flagship for food-photo-heavy venues. Classic Diner is the simple text board that most small restaurants actually want.

[ BUILD ]

Phase 05

Duration: Weeks 10–14

Status: BUILD

Milestone: Can sell to restaurants and cafes

## Photo Grid Layout

- Purpose

QSR, Chinese, Vietnamese, Mexican restaurants · photo-led menu board
- Grid density options

2×2 (4 items) · 3×2 (6 items) · 4×2 (8 items) · 3×3 (9 items)
- Food photo cards

Image (Azure Blob CDN) with gradient fallback · name, desc, price
- Bestseller ribbon

★ POPULAR badge top-left of card · driven by admin toggle
- Sold-out overlay

Card dims with SOLD OUT overlay · auto-restores at midnight
- Happy hour pricing

HH price shown in amber · regular price struck through · only when active
- Multi-screen overflow

Items 1–6 on screen 1 · items 7–12 on screen 2 · self-selecting by position

## Classic Diner Layout

- Purpose

Diners, cafes, sandwich shops · text-only · high legibility at distance
- White background

Inverts the dark theme · warm cream background · dark ink typography

|                          |                                                                       |
|--------------------------|-----------------------------------------------------------------------|
| ■ Multi-column text grid | 2 or 3 columns · sections side by side · familiar laminated menu feel |
| ■ Category headers       | Bold Playfair Display section titles · thin rule below each           |
| ■ Daily special banner   | Full-width strip at bottom · 'Soup of the Day: Tomato Bisque · \$5'   |
| ■ Price alignment        | Right-aligned prices with dot leaders · clean, readable at any size   |

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| Theme Builder — Basic            |                                                                            |
| ■ Available on Starter and above | Basic colour customisation available to all paying customers               |
| ■ Background colour picker       | 6 quick swatches + full colour picker · dark tones only enforced for Diner |
| ■ Accent colour                  | Single brand colour applied to prices, headers, highlights                 |
| ■ Venue name font                | 3 options at Starter: DM Sans, Playfair Display, Syne                      |
| ■ Live preview                   | Admin sees exactly what the TV shows · updates on every interaction        |
| ■ Push to all screens            | One button · SignalR broadcasts theme to every connected screen instantly  |

|                                |                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------|
| Player Evolution — Phase 05    |                                                                                                 |
| ■ Layout switch added          | <code>board.layout === 'photo_grid' → else</code>                                               |
| ■ Azure Blob CDN image loading | Item images served from CDN · lazy loaded with gradient placeholder while loading               |
| ■ Overflow position logic      | <code>screenPos</code> from screen registration · <code>start = (pos-1) × itemsPerScreen</code> |
| ■ Service Worker caching       | Full content payload cached · photos cached separately · offline resilient                      |

06

# Display Layouts — Bars & Restaurants

The Neon Chalkboard is the visual centrepiece of the platform — the display that makes customers stop and look. It is the primary reason a bar chooses Vennu over a generic signage tool. The full Theme Builder unlocks complete visual customisation for Pro venues.

[ BUILD ]

Phase 06

Duration: Weeks 13–17

Status: BUILD

Milestone: Can sell to bars and upscale restaurants

## Neon Chalkboard Layout

- |                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| ■ Chalk board texture          | SVG fractal noise overlay on near-black background · authentic chalk surface                |
| ■ Multi-layer neon text-shadow | 8-layer CSS shadow stack · white core → coloured glow → wide ambient · true tube-light feel |
| ■ Neon flicker animation       | CSS keyframes on venue title · opacity dips at irregular intervals · realistic              |
| ■ Glow breathe animation       | Slow brightness pulse on all neon elements · feels alive, not static                        |
| ■ Chalk draw-in animation      | Items animate in left-to-right with clip-path · like someone writing on the board           |
| ■ Scanline overlay             | Repeating-linear-gradient · subtle TV screen texture across the whole board                 |
| ■ Neon frame + corner brackets | Border with glow shadow · corner bracket accents · frames the board like a real sign        |
| ■ Section neon dividers        | Gradient horizontal rules with glow · matches section colour                                |

## Theme Builder — Full (Pro tier)

**■ 5 built-in preset themes**

Bar Classic · Violet Lounge · Hot Summer · Ocean Dive · Rose Gold

**■ Title colour + glow pickers**

Separate pickers · glow should be a darker shade of colour for depth

**■ Section colours — one per column**

Each menu column gets its own neon ink colour

**■ Glow intensity slider**

0.2× subtle chalk marker all the way to 2.0× full electric overload

**■ Venue name font — 6 options**

Pacifico · Lobster · Righteous · Fredoka One · Bungee · Permanent Marker

**■ Menu items font — 4 options**

Caveat · Kalam · Patrick Hand · Permanent Marker · shown in actual item text

**■ Board background**

Colour picker + 6 quick swatches for common dark board tones

**■ Full Noto font preloading**

SC · KR · JP · Arabic preloaded in display app · prevents flash of unstyled CJK

**Split Layout (Pro tier)****■ Left half — hero photos**

2–4 bestseller photos · large format · strong visual impact

**■ Right half — full text menu**

Complete menu in text · all items · allergen badges · pricing

**■ Responsive split ratio**

Admin can adjust 40/60 or 50/50 split via slider

**■ Purpose**

Mid-range casual restaurants wanting visual appeal and full menu information

**Daily Special Hero Layout (Pro tier)****■ One large featured item**

Full-screen photo + name + description + price · maximum visual impact

**■ Rotating secondary strip**

3–4 smaller items shown below · auto-rotates every 8 seconds

**■ 'Today Only' badge**

Amber banner on the hero item · creates scarcity and urgency

■ Best for

Restaurants that push a new special every day · replaces paper A-frame signs



## 07

## Scheduling Engine

The scheduling engine eliminates the most common daily pain point: manually switching the menu board. Set it once, it runs forever. A Vietnamese restaurant sets up Lunch at 11am and Dinner at 5pm — they never touch the board again.

[ BUILD ]

Phase 07

Duration: Weeks 15–18

Status: BUILD

**Milestone: Menus run themselves — zero daily effort**

### Meal Period Auto-Switch

- |                                   |                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------|
| ■ <b>Period definitions</b>       | Breakfast · Lunch · Afternoon · Dinner · Late Night · each with start/end time               |
| ■ <b>Day-of-week control</b>      | Each period active on configurable days · e.g. Breakfast only Mon–Fri                        |
| ■ <b>Layout per period</b>        | Each period can show a different layout, menu, or theme                                      |
| ■ <b>IHostedService evaluator</b> | Runs every 60s · compares current time to schedules · pushes ContentUpdated if change needed |
| ■ <b>Venue timezone aware</b>     | Schedules evaluated in venue's local timezone · stored as TimeOnly in Azure SQL              |
| ■ <b>Zero staff action</b>        | Once configured, no one needs to touch the board between periods ever again                  |

### Happy Hour Scheduler

- |                              |                                                                             |
|------------------------------|-----------------------------------------------------------------------------|
| ■ <b>Time window config</b>  | Start time + end time + active days · e.g. 4pm–7pm Mon–Fri                  |
| ■ <b>Per-item HH pricing</b> | Each menu item has hhPrice field · shown automatically during active window |
| ■ <b>Auto-activation</b>     | ScheduleEvaluator pushes isHappyHour: true at start time · false at end     |

**■ Manual override**

Admin can force-activate or force-end happy hour regardless of schedule

**■ HH banner on display**

Pulsing 'HAPPY HOUR · 4PM–7PM · MON–FRI' banner shown on board during window

**■ Countdown timer widget**

'Happy Hour ends in 47 min' · proven to drive last-minute orders

**Playlist Rotation****■ Multiple slides per screen**

Screen rotates through slides · configurable dwell time per slide

**■ Slide types**

Menu board · Daily special · Event promo · Image · Custom HTML

**■ Schedule per slide**

Each slide can have its own time window · only shows during configured period

**■ Drag-to-reorder**

Admin drags slides to change rotation order · live preview updates

**Emergency Broadcast****■ One button, all screens**

'Cash only tonight' · 'Kitchen closing in 30 mins' · appears instantly

**■ Full-screen override**

Broadcast takes over entire display · replaces current layout

**■ Auto-expire**

Set a duration · board returns to normal automatically

**■ Venue-wide or screen-specific**

Push to all screens or target individual screens

**Date-Range Promotions****■ Start + end date**

Seasonal specials with defined dates · auto-activates and auto-expires

**■ Christmas menu**

Activates Dec 1 · reverts to regular menu Dec 27 · zero manual intervention

■ Event-specific layouts

Valentine's Day menu · Super Bowl specials · St Patrick's Day board

08

# Tap List Boards — Breweries & Bars

Three distinct tap list board styles covering the full range of brewery and bar aesthetics. Each is modelled on real-world boards seen in actual venues. Tap list is also a standalone data model with beer-specific fields — not a menu with renamed items.

[ BUILD ]

Phase 08

Duration: Weeks 17–21

Status: BUILD

Milestone: Can sell to breweries and taprooms

## Tap List Data Model

|                           |                                                                                 |
|---------------------------|---------------------------------------------------------------------------------|
| ■ TapItem entity          | Separate from MenuItem · name, style, ABV, IBU, description, price              |
| ■ Glass colour field      | Hex colour for the beer glass SVG illustration · dark for stout, amber for IPA  |
| ■ Name colour field       | Each tap gets its own neon colour for the Tap Strips board                      |
| ■ Brewery + origin fields | Brewery name, city, country · shown on Digital Board layout                     |
| ■ On Tap / Off toggle     | Instant 86 for kegs that blow · board updates in <500ms via SignalR             |
| ■ ABV + IBU display       | Both shown on Digital Board · ABV only on Tap Strips · neither on Classic Chalk |

## Classic Chalkboard Drinks Board

|                    |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| ■ Purpose          | Cocktail bars and bars with a fixed beer list · category pricing model                |
| ■ Drinks title     | Large Pacifico-style header with blue neon glow · authentic chalk aesthetic           |
| ■ Category pricing | All cocktails \$10.95 · Import Beer \$4.00 · Domestic Beer \$3.00 · in bordered boxes |

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| ■ Two-column cocktail list | Items in warm gold chalk ink · two columns · Caveat font            |
| ■ Beer sub-sections        | Import and Domestic beers as named lists · bullet-dot separators    |
| ■ Chalk art illustrations  | SVG cocktail glasses and beer bottle drawn in chalk style · centred |

|                          |                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------|
| Tap Strips Board         |                                                                                   |
| ■ Purpose                | Taprooms and craft breweries · each tap gets its own strip · brewery-authentic    |
| ■ 3-column grid          | Strips arranged in a grid · each strip is a dark panel · tap number top-right     |
| ■ Hand-lettered name     | Each tap name rendered in a rotating font (Permanent Marker, Bungee, Pacifico...) |
| ■ Name glow colour       | Each tap has its own neon colour · matches the beer character                     |
| ■ Style + ABV below name | Caveat font · smaller · muted colour · supports the name not competing with it    |
| ■ Draw-in animation      | Strips animate in sequentially · left to right · top to bottom                    |

|                                |                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------|
| Digital Tap Board              |                                                                                  |
| ■ Purpose                      | Modern taprooms wanting a polished digital look · direct Taplist.io competitor   |
| ■ Wood texture background      | SVG fractal noise over warm dark brown · authentic bar atmosphere                |
| ■ Beer glass SVG illustrations | Hand-drawn pint glass SVG · liquid colour matches glassColor field · foam on top |
| ■ Two-column card grid         | 2 columns × 3 rows = 6 taps · each card shows full detail                        |
| ■ Full description text        | Tasting notes shown in full · 2-line clamp · enough to intrigue                  |

|                                |                                                                      |
|--------------------------------|----------------------------------------------------------------------|
| ■ <b>ABV + IBU displayed</b>   | Both metrics shown · appeals to craft beer enthusiasts who read them |
| ■ <b>'Now Brewing' callout</b> | Coming-soon taps shown with special badge · builds anticipation      |

| Screen Registration — Phase 08 Update |                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------|
| ■ <b>Pairing code flow added</b>      | TV loads <code>vennu.app/pair</code> · shows 6-digit code · admin enters code · screen linked        |
| ■ <b>TV polling endpoint</b>          | <code>GET /api/screens/pairing/{code}/status</code> · TV polls every 3s until <code>isClaimed</code> |
| ■ <b>Redirect on claim</b>            | TV auto-redirects to <code>/display/{screenId}</code> when admin claims the code                     |
| ■ <b>Code expiry</b>                  | 10-minute expiry · regenerates automatically · prevents stale codes                                  |
| ■ <b>No URL typing on TV</b>          | Customer never has to type a long URL on a TV remote again                                           |

## 09

# Upgrade Prompts & Billing UX

The in-product upgrade experience that turns free trials into paying customers and paying customers into higher tiers. Every prompt is non-invasive — customers see what they're missing without their current workflow being interrupted.

[ BUILD ]

Phase 09

Duration: Weeks 19–23

Status: BUILD

**Milestone: Self-serve  
upgrade funnel  
generating revenue**

## Core Principles

- |                                           |                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------|
| ■ <b>Show benefit, not tier name</b>      | 'Items sell out → board updates in seconds' · not 'Upgrade to Pro for POS'     |
| ■ <b>Never block a workflow</b>           | Locked features are visible and informative · never an error or dead end       |
| ■ <b>One prompt per screen maximum</b>    | Never show more than one upgrade suggestion at a time · prevents alert fatigue |
| ■ <b>All prompts are dismissible</b>      | Per-session memory · dismissed hints never reappear in the same session        |
| ■ <b>Upgrade is always one click away</b> | Modal → single CTA → Stripe checkout · never more than two taps                |

## Tier Badge

- |                              |                                                                               |
|------------------------------|-------------------------------------------------------------------------------|
| ■ <b>Small coloured pill</b> | e.g. 'PRO' in amber beside any locked feature · informational, never alarming |
| ■ <b>Used on</b>             | Nav items · section headers · individual form fields · layout cards           |
| ■ <b>Colour matches tier</b> | Starter=slate · Restaurant Starter=green · Pro=amber · Business=purple        |

### Locked Nav Items

- |                                |                                                                    |
|--------------------------------|--------------------------------------------------------------------|
| ■ Visible at 50% opacity       | Feature exists in the nav · customers know it's there · not hidden |
| ■ Tier badge beside label      | Clear at a glance which tier unlocks it                            |
| ■ Clicking opens upgrade modal | Not an error · not ignored · a helpful conversation starts         |

### Locked Section Preview

- |                         |                                                                      |
|-------------------------|----------------------------------------------------------------------|
| ■ Blurred glimpse       | 0.3px blur on a mockup of the feature · tantalising, not frustrating |
| ■ One benefit sentence  | 'Prices switch automatically at 4pm — no staff needed'               |
| ■ Soft CTA              | Muted button · 'Unlock with Pro · \$89/mo' · not aggressive          |
| ■ No blocking behaviour | Customer can still use every unlocked feature normally               |

### Inline Feature Hint

- |                         |                                                                             |
|-------------------------|-----------------------------------------------------------------------------|
| ■ Contextual placement  | Appears at bottom of relevant panel · e.g. POS hint on menu editor tab      |
| ■ Amber left accent bar | Subtle visual anchor · amber shimmer animation · feels like a tip not an ad |
| ■ One per panel maximum | Only the most relevant locked feature shown · not a list                    |
| ■ Dismiss button        | × closes it for the session · never nags                                    |

### Sidebar Nudge

- |                                   |                                                    |
|-----------------------------------|----------------------------------------------------|
| ■ Bottom of sidebar               | Quiet, ambient · never in the way of navigation    |
| ■ Rotates through locked features | 7-second intervals · dots show how many are queued |



**■ Per-feature dismiss**

Dismissing one moves to the next · never shows dismissed features again

**■ One sentence + one link**

'Your team can update the board from anywhere · See how →'

**Upgrade Modal — Bottom Sheet****■ Slides up from bottom**

Native-feeling · not a pop-up · backdrop click dismisses

**■ Feature name + benefit**

What this specific feature does · concrete and specific

**■ All tier features as pills**

Every feature at that tier shown · makes total value tangible

**■ Current tier shown**

'You're on Restaurant Starter' · context for the upgrade decision

**■ Single CTA**

'Upgrade to Pro — \$89/mo' · one button · Stripe checkout

**■ 'Maybe later' text link**

Easy non-guilt exit · customer remembers the feature exists

## 10

## POS Integration

Real-time POS sync is the single most powerful differentiator in the platform. When an item sells out at the register, the board updates in under 500ms. No generic signage tool does this well. It is the clearest reason a restaurant chooses Vennu over ScreenCloud, Yodeck, or OptiSigns.

[ BUILD ]

Phase 10

Duration: Months 5–8

Status: BUILD

**Milestone: Beats every generic signage competitor**

### Square — Build First

- |                                  |                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------|
| ■ Why first                      | Best developer API · free sandbox · 4M+ restaurant customers · .NET SDK available |
| ■ OAuth 2.0 connect flow         | 'Connect Square' in admin · one-click auth · access token stored in Azure SQL     |
| ■ Catalog API sync on connect    | Full menu auto-imported from Square · zero manual entry for the customer          |
| ■ Inventory webhook receiver     | POST /webhooks/square · item sells out → board grays out in <500ms                |
| ■ Price sync                     | Price change in Square register → board updates instantly via SignalR             |
| ■ Async processing               | Return HTTP 200 immediately · process in background Task · never times out        |
| ■ Square App Marketplace listing | Free distribution channel to 4M+ restaurant customers · major growth lever        |

### Toast — Build Second

- |                           |                                                                            |
|---------------------------|----------------------------------------------------------------------------|
| ■ Why second              | Dominates full-service restaurant market · Pro and Business tier customers |
| ■ Toast developer sandbox | dev.toasttab.com · free sandbox for development and testing                |

|                              |                                                                             |
|------------------------------|-----------------------------------------------------------------------------|
| ■ Menu availability endpoint | Read-only access — the 86 signal from full-service restaurant POS           |
| ■ Webhook registration       | Not self-serve · submit production URL to Toast contact · requires approval |
| ■ Hourly polling fallback    | Toast recommends polling once per hour as resilience if webhook missed      |
| ■ GUID-based deduplication   | Idempotent processing · safe to receive the same event twice                |

### Clover — Build Third

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| ■ Why third                 | Mid-market gap between Square (SMB) and Toast (enterprise)                     |
| ■ Clover REST API + OAuth   | GET /v3/merchants/{merchantId}/items                                           |
| ■ Inventory update webhook  | Subscribe to inventory events via Clover Developer Dashboard                   |
| ■ Same IPosProvider pattern | Third provider slots straight in · no changes to existing Square or Toast code |

### Integration Architecture

|                                |                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------|
| ■ Unified webhook endpoint     | POST /webhooks/{provider} · one controller handles all three POS systems         |
| ■ IPosProvider interface       | Clean .NET abstraction · new POS systems add one class, nothing else changes     |
| ■ ItemAvailabilityChanged push | SignalR pushes to all venue screens in <500ms of POS event firing                |
| ■ Item patch on client         | Player receives event and patches single item in board state · no full re-render |
| ■ Apideck evaluation           | \$300–500/mo unified wrapper · evaluate when 50+ venues need less common POS     |

11

# Multilingual Support

No signage competitor does multilingual well. A Vietnamese pho restaurant, a Mexican taqueria, a Chinese dim sum place — the owner speaks their language and their customers often prefer theirs. One happy Vietnamese restaurant owner tells five others. This feature opens entire communities through word of mouth.

[ PLAN ]

Phase 11

Duration: Months 7–9

Status: PLAN

Milestone: Ethnic restaurant market unlocked

## Admin UI Translation

- react-i18next Industry standard · auto-detects browser language · stored on User entity
- Language switcher in header Flag emoji + language name · instant switch · persists across sessions
- Launch languages — 3 Spanish · Simplified Chinese · Vietnamese · highest-ROI non-English markets in the US
- Phase 2 languages Korean · Portuguese · add after validating initial three
- All UI strings externalised Zero hardcoded English text anywhere in admin · enforced from Phase 04

## Bilingual Display Modes

- Stack mode Primary language large · secondary language smaller below · cleanest option
- Side-by-side mode Two full columns · one per language · good for equal-prominence bilingual venues
- Subtitle mode Primary large · translation as small italic subtitle · minimal extra space

**■ Per-venue setting**

Admin chooses mode · stored on Venue entity · applied to all layouts

**■ Font pairing**

English stays in Caveat/Kalam · CJK/Arabic uses matching-weight Noto font

**AI Bulk Translation — Claude API****■ One-click translate entire menu**

'Translate to Chinese' button · Claude translates all items in ~15 seconds

**■ Cost**

~\$0.30 to translate 50 items into 3 languages · essentially free

**■ Context-aware**

Claude understands restaurant terminology · 'Kung Pao' → '■' not a literal translation

**■ Review table**

Owner reviews all translations · edit any inline · mark as reviewed

**■ IsAutoTranslated flag**

DB flag distinguishes AI translations from owner-entered · for quality tracking

**RTL Support****■ Arabic + Hebrew**

dir=rtl applied to RTL language content · layout mirrors appropriately

**■ Noto Sans Arabic preloaded**

Loaded in display app regardless of current settings · instant render

**■ Admin RTL mode**

Admin UI can also switch to RTL for Arabic/Hebrew-speaking owners

**Screen Registration — Phase 11 Update****■ Language on pairing screen**

vennu.app/pair shown in browser language automatically · Chinese owner sees Chinese

**■ Venue language inherited**

New screens inherit venue's primary and secondary language on first load



1  
2

# Staff Mobile App

The single most-requested feature after launch. A bar manager who can't find the laptop at 6pm on a Friday will cancel their subscription. The one who can update the board from their iPhone will refer Vennu to every venue they know. This app is the primary retention driver for the Pro tier.

[ PLAN ]

Phase 12

Duration: Months 9–11

Status: PLAN

Milestone: Pro tier becomes sticky

## React Native — iOS + Android

- One codebaseReact Native · submitted to App Store and Google Play · included in Pro and above
- Shared APISame .NET endpoints as web admin · no new backend work required
- Biometric authFace ID / Touch ID · fast login for operators who check it mid-service
- Push notificationsFirebase Cloud Messaging · screen offline alerts · keg blow · daily special reminder

## Core Actions

- Quick 86 toggleMark item unavailable in 2 taps from the kitchen · most-used feature post-launch
- Daily special pushText entry → live on all screens in under 10 seconds
- Happy hour overrideActivate or deactivate happy hour instantly regardless of schedule
- Emergency broadcast'Cash only tonight' from the phone · all screens update immediately

## Screen Health

|                              |                                                                    |
|------------------------------|--------------------------------------------------------------------|
| ■ All screens at a glance    | Online / offline status · last-seen · quick push button            |
| ■ Offline alert notification | Push alert the moment any screen goes offline during service hours |
| ■ Remote content refresh     | Force a screen to reload its content with one tap                  |

|                         |                                                                       |
|-------------------------|-----------------------------------------------------------------------|
| Brewery-Specific        |                                                                       |
| ■ Keg blow notification | Staff marks keg empty → board removes tap + manager push notification |
| ■ Tap list quick-edit   | Change ABV, price, or description directly from the phone             |
| ■ 'Now Pouring' update  | Mark a new keg as live · board updates the tap strip immediately      |



1  
3

# TV Apps & Platform Distribution

Get Vennu into the TV app stores so customers can install it like any other app. The display player code is unchanged — these are thin native wrappers around the existing React SPA. The pairing code flow handles setup with zero URL typing.

[ PLAN ]

Phase 13

Duration: Months 11–15

Status: PLAN

**Milestone: App store presence on all major TV platforms**

## Android TV + Amazon Fire TV

### ■ Why first

Fire TV Stick is the most common hospitality hardware · Android TV covers the rest

### ■ Kotlin WebView wrapper

Thin native shell · `loadUrl('vennu.app/display/{screenId}')` · one APK covers both

### ■ Pairing code on first launch

App shows 6-digit code automatically · admin claims it · screen linked in 30 seconds

### ■ Kiosk / pinned mode

Prevents exit without PIN · Fire TV settings locked · always-on display

### ■ Auto-launch on TV startup

Android boot receiver · app starts automatically when TV powers on

### ■ Google Play for TV submission

'Vennu' searchable in TV app store · customers install it like Netflix

### ■ Amazon Appstore submission

Same APK · separate submission · covers all Fire Stick variants

## Samsung Tizen

### ■ Why important

Samsung dominates commercial display market · app store listing removes all setup friction

**■ Tizen web app package**

React SPA + config.xml manifest · submitted to Samsung Smart Signage Platform

**■ Zero player code changes**

Same display app · Tizen just packages and hosts it · pairing code flow works identically

**■ Samsung B2B program**

Enables pre-installation on shipped HaaS units

**LG webOS****■ IPK package**

Same React SPA · different build target · submitted to LG Smart Signage Solution

**■ webOS simulator**

Free LG developer tool · test without physical hardware

**■ Coverage**

LG is the second largest commercial display brand globally

**Screen Registration — Phase 13 Update****■ TV app shows pairing code on launch**

Fullscreen display of 6-digit code + [vennu.app/pair](#) instruction

**■ Auto-redirect after claim**

TV transitions from pairing screen to content display seamlessly

**■ Platform field on Screen entity**

Stores 'chrome', 'android\_tv', 'fire\_tv', 'tizen', 'webos' · shown in super admin

**■ App version tracking**

Player reports app version in heartbeat · super admin shows outdated screens

**HaaS Pre-Registration****■ Screens arrive pre-configured**

You create screen records before shipping · venue's content loaded at factory

**■ Plug in and it works**

Customer powers on the TV · correct menu already showing · zero setup required

**■ Card in the box**

'Your screens are already set up. Plug in, power on, done.'

■ Premium HaaS experience

No pairing code needed · screens registered to venue before delivery

## 14

## Super Admin CRM

Until you have ~30 customers you are the super admin — manage tiers directly in Azure SQL. Build this when managing venues manually becomes risky and time-consuming. The super admin is the operating layer over the tier system built in Phase 03.

[ PLAN ]

Phase 14

Duration: Months 13–16

Status: PLAN

Milestone: Internal ops at scale — ~30 customers

### Revenue Dashboard

- **MRR + ARR** Live from Stripe · month-over-month trend · breakdown by tier
- **Active venues** Count + list · trialing separately · churned in last 30 days
- **Avg revenue per venue** MRR ÷ active venues · track over time
- **Recent events feed** Upgrades · new signups · overrides applied · churn · in chronological order
- **Screen health map** Every screen across every venue as a coloured dot · hover for venue + screen name

### Tier Manager

- **Create / edit / clone / archive** Full CRUD on tiers · no code deployment needed · live immediately
- **Feature toggle per tier** Check / uncheck any feature flag for any tier · save applies instantly
- **Enable All / Clear All shortcuts** Bulk toggle for quick tier configuration
- **Stripe product ID field** Links tier to Stripe product · kept in sync on save
- **Public / private flag** Private tiers for custom enterprise deals · not shown on pricing page

## Feature Matrix

- **All features × all tiers as a grid** One cell = one checkbox · click to toggle · amber highlight on unsaved changes
- **Category grouping** Display · Scheduling · Language · Mobile · POS · AI · Analytics · Enterprise
- **Save all changes at once** One save button applies all pending changes simultaneously
- **Audit trail** Every feature change logged with admin ID and timestamp

## Venue CRM

- **Search + filter** By name · tier · status (active, trialing, cancelled) · health
- **Venue table** Name · type · tier · MRR · screen count · last active · override count · health
- **Tier switcher** Change any venue's tier in one click · Stripe updated automatically
- **Override panel** Add feature unlock or block · reason text · optional expiry · takes effect immediately
- **Effective features view** Shows each feature's source (tier vs override) · full transparency per venue
- **Support context** See exactly what a customer has access to before responding to a support request

15

# AI Features

AI that lowers the skill floor for every venue owner. A restaurant operator who has never written a menu description or designed a layout gets professional results in seconds. Powered by Claude API — ~\$0.01–0.03 per generation.

[ PLAN ]

Phase 15

Duration: Months 14–18

Status: PLAN

Milestone: Premium tier upsell driver

## AI Menu Content — Claude API

|                               |                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| ■ Menu description writer     | 'grilled salmon, lemon butter' → 'Pan-seared Atlantic salmon with house-made lemon beurre blanc' |
| ■ Bulk description generation | Generate descriptions for every item with no descriptions · one click · reviewed inline          |
| ■ AI translation              | Bulk translate entire menu into 3 languages for ~\$0.30 · covered in Phase 11                    |
| ■ Smart naming suggestions    | Highlight generic item names like 'Chicken Dish' · suggest more appealing alternatives           |
| ■ Allergen detection          | AI reads description and suggests relevant allergen badges · owner confirms                      |

## AI Custom Display Builder (Business tier)

|                                |                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------|
| ■ Plain English input          | 'Dark blue, gold text, logo top centre, beer list left, food right, countdown bottom' |
| ■ Claude generates layout      | Full JSX/CSS generated · saved to Azure Blob · loaded into iframe sandbox             |
| ■ Template variables available | {{venueName}} · {{items}} · {{isHappyHour}} · {{currentTime}} · {{sections}}          |
| ■ iframe sandbox               | sandbox='allow-scripts' · security boundary · SignalR still pushes live updates       |

**■ Saved and reusable**

Generated templates stored · owner regenerates with new prompt anytime

**■ ~\$0.02 per generation**

Business tier only · metered usage visible in super admin

**HTML/CSS Sandbox Editor (Developer tier)****■ Monaco Editor**

VS Code engine in the browser · syntax highlighting · autocomplete

**■ Template variable panel**

Sidebar shows all {{variables}} with live examples

**■ Real-time preview**

Renders with real menu data while editing · no save needed to see result

**■ Not shown to standard owners**

Developer tier feature · separate plan for tech-savvy or agency operators

**Smart Happy Hour Suggester****■ Analyses POS slow periods**

Looks at order volume by hour across past 30 days

**■ Recommends optimal windows**

'Your slowest period is 3pm–5pm Tue–Thu · consider a happy hour here'

**■ One-click apply**

Accept suggestion → schedule configured automatically

**Photo Background Generator****■ Upload plain food photo**

Customer uploads a basic photo from their phone

**■ AI generates styled background**

Matches venue brand colours · appropriate for the display layout

**■ Output saved to Azure Blob**

Ready to use on photo grid cards immediately

## 16

## Analytics & Smart Features

Analytics needs real usage data and POS history to be meaningful — it belongs at the end of the roadmap, not the beginning. Smart features like dynamic pricing and A/B testing require a solid data foundation and confident customers to be valuable.

[ PLAN ]

Phase 16

Duration: Months 17–22

Status: PLAN

**Milestone: Data-driven venues — clear ROI proof**

### Analytics Dashboard

- |                                     |                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------|
| ■ <b>Screen uptime tracking</b>     | Heartbeat pings logged · uptime % per screen per week · downtime alerts          |
| ■ <b>Content impression logging</b> | Which items displayed · when · for how long · total impressions per item per day |
| ■ <b>POS sales correlation</b>      | Display position vs. actual sales · powered by Square/Toast integration data     |
| ■ <b>Item performance scoring</b>   | 'Your #2 seller by revenue sits at screen 3 position 8 — consider moving it'     |
| ■ <b>Happy hour ROI report</b>      | Order volume before / during / after happy hour · proves financial value         |
| ■ <b>Venue comparison</b>           | For multi-location Business tier · compare performance across branches           |

### A/B Layout Testing

- |                                              |                                                                             |
|----------------------------------------------|-----------------------------------------------------------------------------|
| ■ <b>Test two layouts against each other</b> | Layout A shown Monday/Wednesday · Layout B shown Tuesday/Thursday           |
| ■ <b>Metric</b>                              | Average order value from POS · items sold per session                       |
| ■ <b>Automatic winner selection</b>          | After 2 weeks system identifies winner · prompts admin to make it permanent |



**■ Item position testing**

Test item A in position 1 vs position 4 · measure sales impact

**Dynamic Pricing****■ Time-of-day pricing**

Lunch specials at \$9.99 auto-activate at 11am · revert to full price at 3pm

**■ Demand-based pricing**

Slower periods get discounted prices automatically · no staff action

**■ Rules engine**

Admin sets conditions: 'If day is weekday AND hour &lt; 5pm THEN use lunch price'

**■ Business tier only**

Yield management tool for sophisticated operators

**Smart Context Features****■ Foot traffic mode**

Google Maps Popular Times API → simplified board during peak rush hours

**■ Weather-triggered promotions**

Temperature &gt; 85°F → cold drink special auto-activates

**■ Social proof ticker**

Google and Yelp review rotation at point of decision

**■ Upsell prompt engine**

'Pairs well with our Hazy IPA' shown beneath featured items

**■ Loyalty QR integration**

Square Loyalty and Toast Loyalty sync · points balance QR on screen

**■ Live sports ticker**

Footer bar for sports bars · keeps customers watching and ordering

**Webhook & Automation****■ Outbound webhooks**

Vennu fires events to customer's own systems on content changes

**■ Zapier integration**

Connect to anything · Google Sheets price sync · weather API · Slack alerts

■ Google Sheets price sync

Owner updates a price in Google Sheets · Vennu board updates automatically

# Revenue Model

## Software-Only Pricing

| Tier               | Price    | Screens   | Key Features                                                             |
|--------------------|----------|-----------|--------------------------------------------------------------------------|
| Starter            | \$39/mo  | 2         | Photo Grid · Classic Diner · basic scheduling · allergen badges          |
| Restaurant Starter | \$49/mo  | 1         | Meal periods · bilingual · AI translation (1 lang) · Quick Update · comb |
| Pro                | \$89/mo  | 6         | All layouts · happy hour · POS · staff app · video wall · full analytics |
| Business           | \$179/mo | Unlimited | Everything + AI custom builder + multi-location + white-label            |

## Hardware-as-a-Service Bundles

| Bundle      | Monthly  | Term  | Includes                      | Post-Contract       |
|-------------|----------|-------|-------------------------------|---------------------|
| Starter Kit | \$89/mo  | 18-mo | 1 screen + Fire Stick         | → \$39/mo Starter   |
| Bar Pack    | \$159/mo | 24-mo | 2 screens + mounting + sticks | → \$89/mo Pro       |
| Full House  | \$249/mo | 36-mo | 4 screens + mounting + setup  | → \$179/mo Business |

*Hardware ownership transfers at contract end — no return required. Post-contract, Stripe auto-transitions the venue to the equivalent software tier. HaaS LTV over 5 years: Full House = ~\$13,000 vs ~\$1,600 software-only (8× more valuable).*

## MRR Milestones — Pro @ \$89/mo

| Customers | MRR     | Annual   | Significance                                            |
|-----------|---------|----------|---------------------------------------------------------|
| 4         | \$356   | \$4,272  | Infrastructure break-even · Azure + Stripe fees covered |
| 10        | \$890   | \$10,680 | Ramen profitable                                        |
| 65        | \$5,785 | \$69,420 | Break-even including \$5,000/mo founder salary          |

|     |          |           |                                 |
|-----|----------|-----------|---------------------------------|
| 100 | \$8,900  | \$106,800 | Healthy single-founder business |
| 200 | \$17,800 | \$213,600 | First hire justified            |
| 500 | \$44,500 | \$534,000 | Series-A territory              |

## Immediate Next Steps

### This Week

- **File Vennu trademark** tess.uspto.gov · Class 042 · do not delay · foundational brand protection
- **Secure venuu.app** Register domain immediately · \$12/yr · also check getvennu.com
- **Set up Azure subscription** Resource group · App Service B1 · Azure SQL Basic · Blob Storage
- **Initialise GitHub repo** .NET 8 Web API + Vite React monorepo · connect GitHub Actions for CI/CD

### Weeks 2–4

- **Scaffold EF Core data models** Include Feature, TierFeature, VenueSubscription, ScreenPairingCode from day 1
- **Build SignalR hub** JoinScreen + ContentUpdated · test real-time push with two browser tabs
- **Set up Stripe** Create product + price objects for all 4 tiers · monthly and annual
- **Build pairing code flow** venuu.app/pair · 6-digit code → admin claims → screen linked in 30s

### Month 2

- **Build tier feature service** HasFeatureAsync(venueId, key) · override first, tier fallback · all future features use this
- **Build admin menu editor** Inline edit, 86 toggle, Quick Update mobile view · tier-aware from first line
- **Build Photo Grid layout** First display layout · real-time update via SignalR · offline Service Worker cache
- **First beta customer** One venue, one screen, \$0 while testing · get real feedback before building more

### Month 3

- |                                     |                                                                    |
|-------------------------------------|--------------------------------------------------------------------|
| → <b>Build Classic Diner layout</b> | Second layout · unlocks cafes and diners as a market               |
| → <b>Build Neon Chalkboard</b>      | Flagship layout · bars and restaurants · theme builder             |
| → <b>Launch scheduling engine</b>   | Meal periods + happy hour · IHostedService evaluator               |
| → <b>First paying customer</b>      | Target \$39–89/mo by end of month 3 · validates willingness to pay |

### Month 4–5

- |                                       |                                                                              |
|---------------------------------------|------------------------------------------------------------------------------|
| → <b>Build tap list boards</b>        | All 3 styles · tap item data model · on-tap toggle via SignalR               |
| → <b>Build upgrade prompts</b>        | Tier badges · locked sections · inline hints · sidebar nudge · upgrade modal |
| → <b>Start Square POS integration</b> | OAuth connect flow · catalog sync · inventory webhook receiver               |
| → <b>Stripe billing live</b>          | Customers can self-serve upgrade · super admin data model ready              |